	Hence the amplitude of the oscillations decreases continuously over time.
7	
(a)(i)	$z = 92 - 42 = 50$
	$x = 235 + 1 - 99 - 134 = 3$
(a)(ii)	energy released $= (8.61 \times 99) + (8.45 \times 134) - (7.59 \times 235)$
	$= 201.04$ $= 201$ MeV (3 sf)
(a)(iii)	The reaction produces more neutrons which can react with Uranium nuclides to produce even more neutrons.
(b)(i)1	It is impossible to predict when a particular nucleus in a sample is going to decay. OR Each nucleus in a sample has the same chance of decaying per unit time.
(b)(i)2	Not affected by external factors such as presence of other nuclei, temperature and pressure.
(b)(ii)	${}^{99}_{42}\text{Mo} \rightarrow {}^{99m}_{43}\text{Tc} + {}^{0}_{-1}\text{e} + \bar{\nu} + \text{energy}$
(b)(iii)	mass defect $= (98.90771u - 98.90641u - 0.00055u)$ $= 7.5 \times 10^{-4} u$
	energy released $= (7.5 \times 10^{-4})(1.66 \times 10^{-27})(3.00 \times 10^{8})^2$ $= 1.1205 \times 10^{-13}$ J

	$= \frac{1.1205 \times 10^{-13}}{10^6 \times 1.60 \times 10^{-19}}$ MeV $= 0.70031$ MeV $= 0.700$ MeV (3 sf)
(c)(i)	Probability per unit time that a nucleus will undergo decay.
(c)(ii)	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{66 \times 60 \times 60} = 2.91729 \times 10^{-6}$ s^{-1}
	$N = \frac{A}{\lambda} = \frac{12.0 \times 10^9}{2.91729 \times 10^{-6}} = 4.11341 \times 10^{15}$
	mass $= 4.11341 \times 10^{15} \times 98.90771 \times 1.66 \times 10^{-27}$ $= 6.75367 \times 10^{-10}$ $= 6.75 \times 10^{-10}$ kg (3 sf)
(c)(iii)	$m = m_0 e^{-\frac{\ln 2}{t_{\frac{1}{2}}} t}$ $= 6.75 \times 10^{-10} e^{-\frac{\ln 2}{\left(\frac{66}{24}\right)} 30}$
	$= 3.51027 \times 10^{-13}$ $= 3.51 \times 10^{-13}$ kg (3 sf)
(d)(i)	Short half-life of 6 hours allows sufficient time to perform scan OR It decays quickly enough to reduce radiation dose. OR γ–radiation is weakly ionising, which minimises tissue damage. OR γ–radiation is highly penetrating, which allows it to be easily detected outside the patient.
(d)(ii)	Decay constant is extremely small, activity inside the patient is negligible.
(d)(iii)	Industrial tracking takes place over long distances / long periods of time. A short half-life would decay to undetectable levels before process is complete.